

OpenPI Comet

Stakeholder · Kanta

OpenPI Comet / BEHAVIOR Challenge / 2nd place

18+3 min

arXiv 2512.10071 · BEHAVIOR Challenge 2025 · smoke-run paper club

System / method pipeline

OpenPI Comet pipeline

π 0.5 backbone · heterogeneous pre-train · RFT flywheel · RH inference

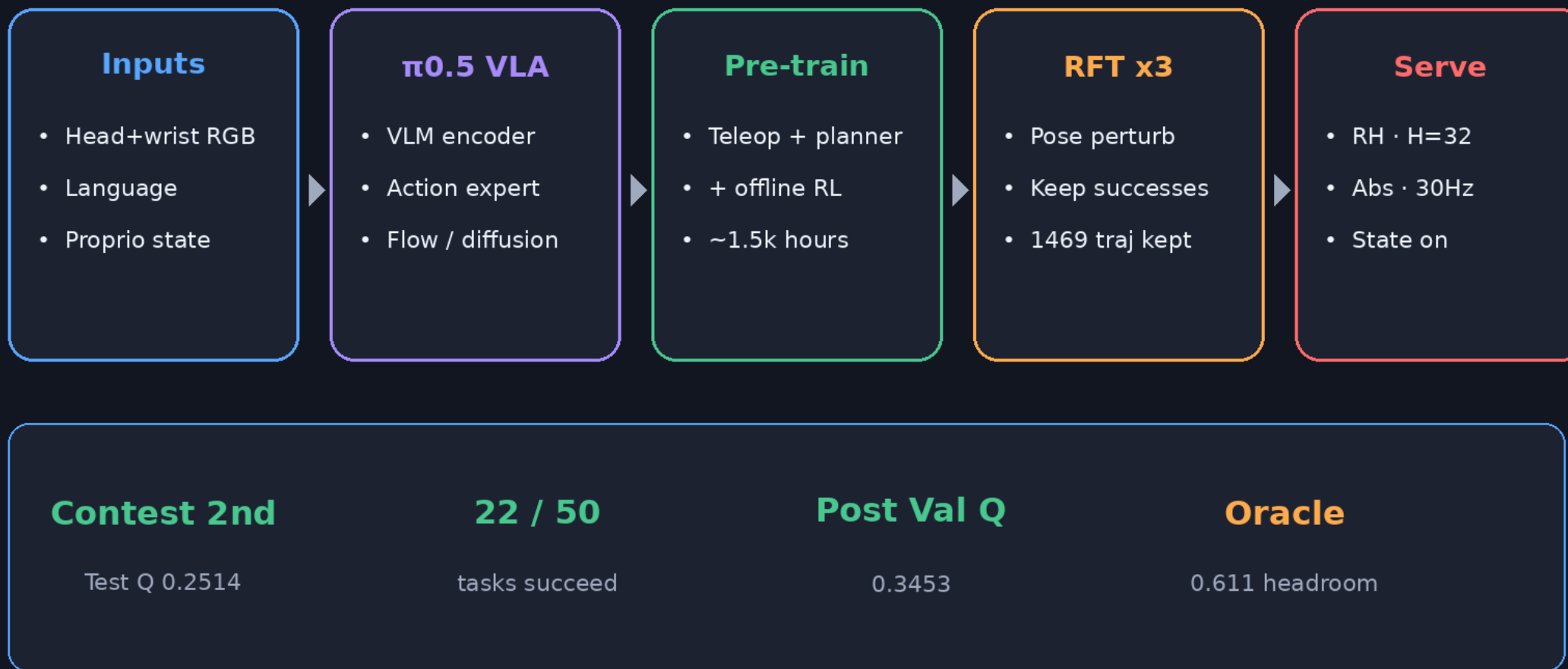
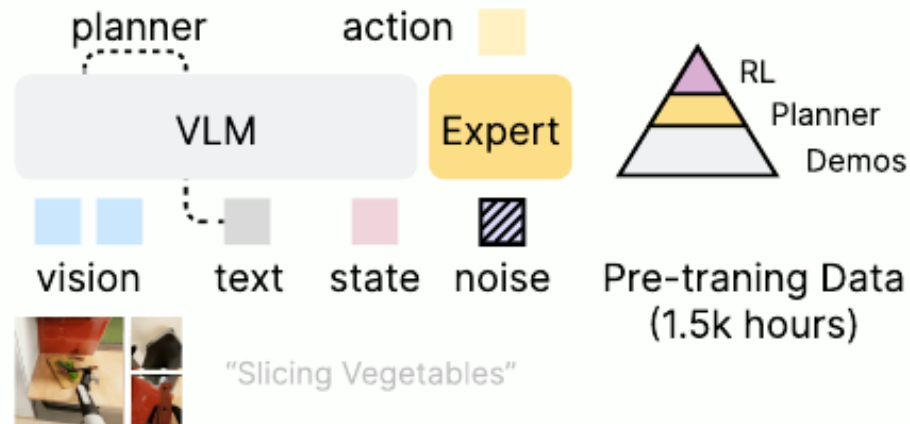


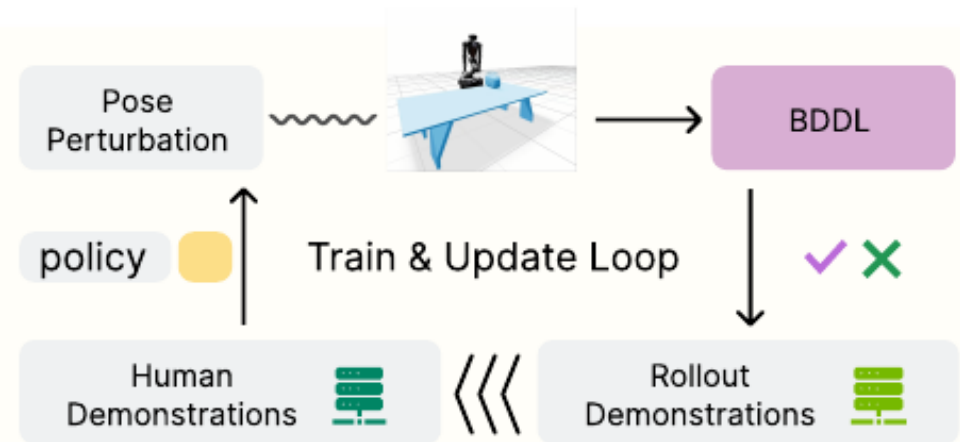
Fig 1 — Pre-train + RFT flywheel (paper)

and online RL rollouts, which together supply rich long-horizon behaviors and diverse manipulation strategies crucial for robust policy learning.

For post-training, we adopt an iterative RFT procedure as illustrated in Figure 1(b). Starting from the official human demonstrations, we introduce random pose perturbations and roll out the pretrained policy under these disturbed initial conditions. Successful episodes are retained as additional training data, progressively forming an offline data flywheel that continually improves the robustness and coverage of the policy.



a) Pre-training Stage



b) Post-training Stage

Figure 1: (a) Pre-training on large-scale heterogeneous data, including 1.1K hours of human demonstrations and ~ 0.4 K hours of additional planner and offline RL trajectories. (b) RFT post-training: perturb initial poses, roll out the policy, and retain successful episodes to iteratively augment the dataset.

Leaderboard — Table 1

Table 1 — 2025 BEHAVIOR Challenge (standard track)

Entry	Val SR	Test SR	Val Q	Test Q
Comet post-challenge	0.1500	—	0.3453	—
#1 Robot Learning Collective	0.1120	0.1240	0.2605	0.2599
#2 Comet (competition)	0.1440	0.1140	0.1830*	0.2514
#3 SimpleAI Robot	0.1400	0.1080	0.1943	0.1591
#4 The North Star	0.1280	0.0760	0.1702	0.1204

* Incomplete Val eval by deadline. Competition rank by Test Q → Comet 2nd (0.2514).

Coverage ladder — pt1 → pt7 → pt10 → pt50

Coverage ladder — pre-train task diversity

Successful-task count rises with coverage (Fig 3)



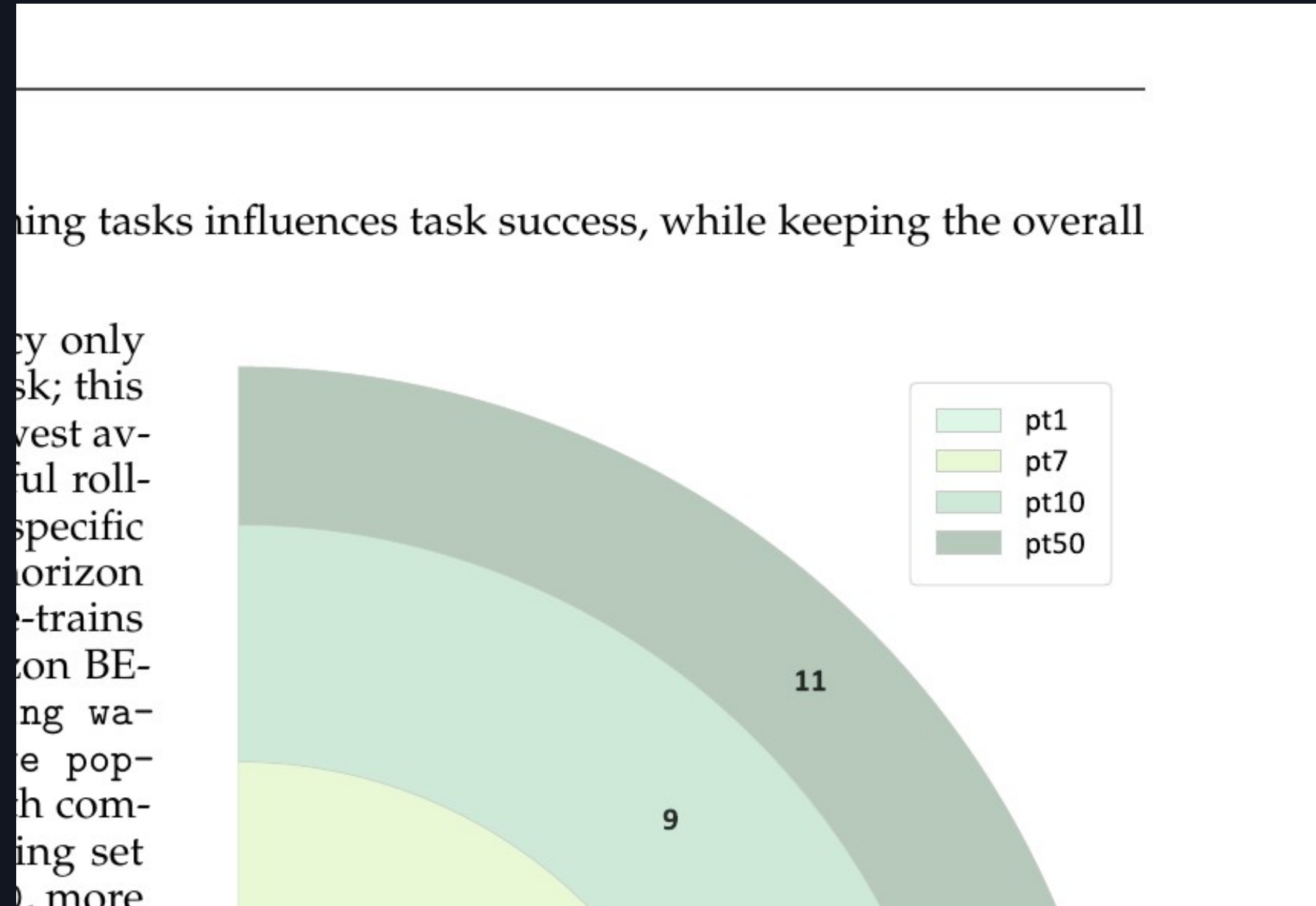
After pt50

Val Q = 0.192

Still only 11/50
tasks show success

→ **need RFT**

Fig 3 — paper coverage figure



RFT post-train — 0.192 → 0.345 · oracle 0.611

RFT post-train story

Rejection Sampling Fine-Tuning · N=3 · ~8500 rollouts/round → 1469 kept



Defaults that matter

Inference / data defaults that matter

On turning_on_radio — wrong knob → ~0% SR

Control

Receding Horizon

Ensemble / Receding Temporal → 0.00

Horizon H

H = 32

H=8 → 0.00 · H=16 → 0.10

Actions

Absolute joints

Delta joints → 0.00

Rate

30 Hz

15 Hz → 0.00

State

Proprio ON

No state → 0.00

Vision

RGB 224 (or 720↑)

High-res → 0.60 (2×)

Qualitative rollouts (Fig 5)

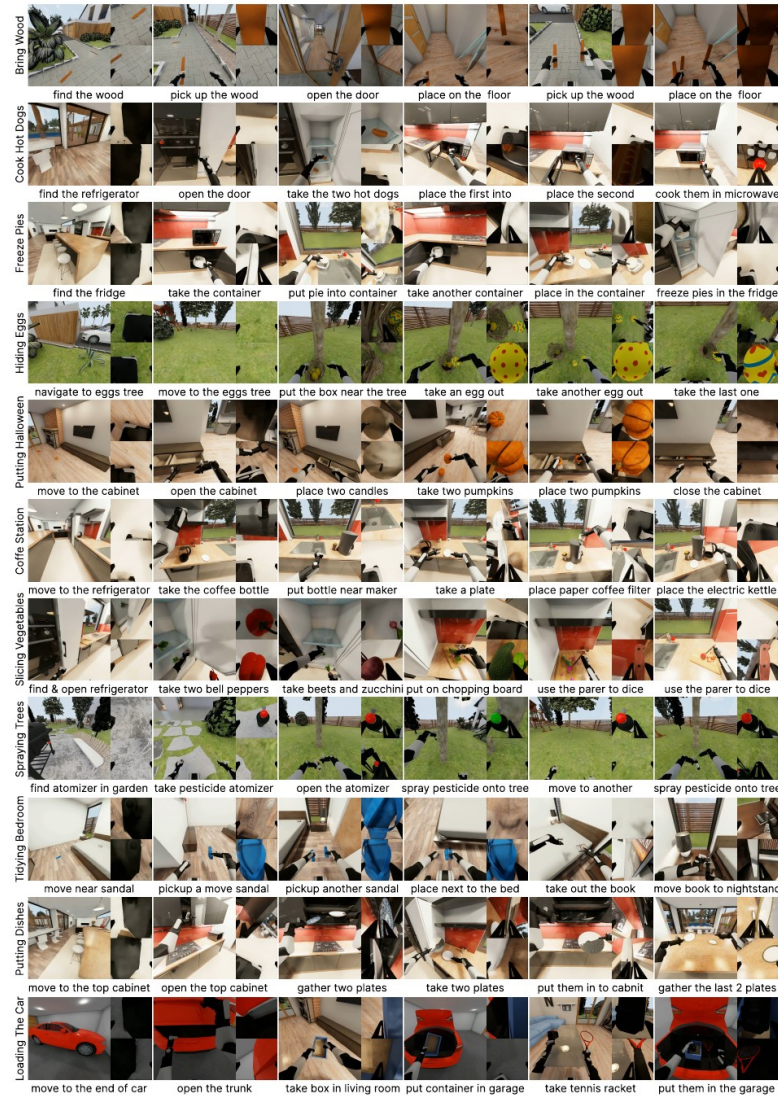


Figure 5: Rollout Examples on BEHAVIOR-1K. Our policy successfully handles long-horizon, multi-stage

Findings + takeaway

18+3

Findings + takeaway

Public VLA + flywheel

$\pi 0.5$ + data mix + RFT + RH
is competitive (2nd place)

Defaults are brittle

Wrong controller / Hz / frame
→ ~0% on easy probe

Headroom remains

0.345 → 0.611 oracle
Hard half of suite still fails

Gap to close

Post-train efficiency +
hard-task curricula

**Public VLA + data flywheel is competitive —
close the hard-task gap next.**

Q&A · timing 18+3